# Handwriting Analysis for Personality Prediction and Career Counseling

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**Abstract**

One unique biometric feature that shows not only motor skills but also sophisticated cognitive and psychological processes is handwriting. Its ability to encode faint behavioural clues makes it a useful tool for personality evaluation. Leveraging both Machine Learning (ML) and Deep Learning (DL) approaches, this work proposes an automated data-driven system for predicting personality traits by handwriting analysis. The approach uses a bilingual dataset consisting of handwritten samples in Hindi and English, therefore allowing cross-linguistic applicability. The system relates important traits including stroke dynamics, spacing, slant, pressure, and curvature to the Big Five personality dimensions: Openness, conscientiousness, extraversion, agreeableness, and neuroticism by means of key feature extraction.

To improve the generalising capacity of the model across various handwriting styles, a hybrid model architecture is presented combining handmade feature engineering with deep neural network-based representations. Additionally included for behaviour inference and contextual feedback were transformer-based models and refined large language models (LLMs) from the Hugging Face. Beyond basic categorisation, the system integrates the predictive capacity of artificial intelligence with human-centric design to provide individualised recommendations, career alignment advice, and behavioural insights.

Moreover, this work investigates how LLMs might offer adaptive personality-aware interactions, thereby supporting useful applications in mental health assistance, job development, and education. High accuracy, stability, and robustness across multilingual data shown by experimental tests validated the efficacy of the suggested method. Promoting the creation of ethical, inclusive, and explainable artificial intelligence systems for personality assessment, this work greatly advances the fields of emotional computing, computational psychology, and intelligent advising systems.

**Keywords: Big Five Personality Traits, Machine Learning, Deep Learning, Handwriting Analysis, Multilingual Dataset, Large Language Models (LLMs), Hugging Face, Personality Prediction.**

## I. Introduction

An vital form of human expression that reflects not only verbal patterns but also psychological and cognitive aspects is handwriting. Researchers have investigated the possibility of handwriting analysis as a nonintrusive approach for personality evaluation over years. Conventional graphology-based techniques have sought to correlate fundamental personality qualities with handwriting features including slant, pressure, spacing, and stroke patterns.

These approaches, meanwhile, sometimes suffer from restricted generalisability, subjectivity, and lack of standardising. Automated handwriting analysis has attracted lot of interest with developments in machine learning (ML) and deep learning (DL). A promising substitute for conventional psychological tests are ML models' capacity to diagnose personality traits with great accuracy and extract significant features from handwriting samples.

Conscientiousness, Openness, Extraversion, Agreeableness, and Neuroticism make up the Big Five Personality Traits model, which provides a generally agreed upon framework for computational personality prediction. Though much has been done, some problems still exist in this field. Many current research have concentrated mostly on

English handwriting reduces their relevance in many linguistic and cultural settings.

Furthermore difficult for feature extraction and categorisation are differences in writing style resulting from outside elements including emotional state, writing surface, and writing tools. Moreover, a major obstacle for personality prediction still is the availability of big-scale well-annotated handwriting datasets.

This work intends to overcome these difficulties by means of an automated personality prediction system grounded on handwriting analysis. The main contributions of this paper consist in The generalisability of the model was improved using a multilingual dataset including handwritten examples from English and Hindi. Advanced machine-learning pipeline including approaches based on deep-learning for categorisation and feature extraction.

Strong techniques for managing handwriting variances increase the model's flexibility in several writing styles. This work aims to improve the accuracy and dependability of handwriting-based personality prediction by using state-of- the-art ML and DL approaches. Apart from improving computational psychology research, the suggested approach has possible uses in domains including recruitment, behavioural analysis, and mental health evaluations.

## II. Literature review

Researchers in many areas including psychology, machine learning, and computer vision have become interested in the developing topic of handwriting analysis for personality prediction. Mostly applying psychological theories like the Big Five Model and HEXACO, some studies have tried to find an association between handwriting qualities and personality traits. To find significant trends from handwritten data, researchers have investigated conventional graphology-based techniques and computer approaches.

Graphological ideas, which propose that particular handwriting traits—such as slant, pressure, letter spacing, and stroke patterns—can expose the underlying personality traits—have the basis of handwriting-based personality prediction. To find such associations, early research mostly depended on statistical analysis and hand feature extraction.

Modern research has turned towards automated feature extraction utilising image processing and neural networks, though, as machine learning and deep learning approaches have emerged. Recent work suggests several approaches for feature extraction, from handcrafted features to deep-learning-based representations. Extensive application of classic machine learning techniques including Support Vector Machines (SVM), Decision Trees, and Random Forest has helped to categorise personality traits depending on acquired handwriting data. Conversely, by learning challenging feature representations from raw handwritten photos, deep learning models—especially Convolutional Neural Networks (CNNs)—have shown amazing performance. Moreover, hybrid models combining deep learning elements with handmade aspects were presented to improve classification accuracy.

Notwithstanding these developments, some important problems still exist. One of the main constraints is the dearth of freely available, large-scale handwriting datasets labelled with personality variables. Variations in writing style resulting from outside variables such emotional state, writing tools, and surface conditions also add noise in the data, therefore affecting the prediction accuracy. Many current research also centre on a particular script (mainly English), therefore restricting the generalisability of the results to other languages and writing systems. Dealing with these constraints calls both domain-adaptive models and better feature-extraction methods as well as strong datasets.

## III. Related works

Several methods have been suggested for personality prediction via handstroke analysis by combining conventional and contemporary computational methods. Early study mostly depended on statistical and heuristic approaches, in which case researchers personally found handwriting characteristics connected to psychological characteristics. With slant, pressure, curvature, and letter size the most often examined properties, feature engineering was absolutely vital in these investigations.

To forecast the personality types, these traits were then fed into classification systems including SVM and k-nearest neighbours (k-NN). Automated feature-extraction techniques have evolved from machine learning and found acceptance. Many research have extracted spatial information from handwriting samples using histograms of orientated gradients (HOG) and Gabor filters then followed by supervised classification algorithms. Other studies have looked at deep learning methods—especially CNNs—to replace human feature extraction. Handwriting photographs have been fed refined pre-trained models including ResNet and VGG-16, which have shown appreciable classification accuracy increase. Additionally showing encouraging results are hybrid models combining deep learning with conventional feature engineering techniques.

To address the difficulties presented by limited labelled datasets, another line of study has investigated semi-supervised learning and domain adaption methods. On limited handwriting datasets, generative adversarial networks (GANs) and transfer learning have been used to improve the model performance. By stressing the most important handwriting patterns guiding categorisation decisions, studies including attention mechanisms have improved the interpretability of handwriting-based personality prediction.

Methodologically, past research have tested several preprocessing approaches like binarization, normalisation, and augmentation to increase handwriting feature extraction. Researchers have also used explainable artificial intelligence models to analyse model predictions, therefore guaranteeing more dependability and openness. To create more complete personality evaluation systems, some research have also incorporated multimodal techniques integrating handwriting analysis with speech recognition and keystroke dynamics.

Notwithstanding these developments, current approaches still suffer with regard to dataset standardising, cross-linguistic generalisability, and handwriting variation resilience. By merging multimodal techniques and handwriting analysis with other behavioural cues like keystroke dynamics and speech patterns for more complete personality assessments, future studies should solve these difficulties. Moreover, using explainable artificial intelligence methods might improve the dependability and openness of handwriting-based personal prediction systems.

## IV. Dataset

This work used a varied collection of 3,416 handwritten samples, methodically arranged according to the Big Five personality traits: Openness, Conscientiousness, Agreeableness, Neuroticism, and Extraversion. This dataset stands out mostly for its bilingual composition—handwriting samples in English and Hindi. This linguistic diversity guarantees more general application of personality prediction across many populations by improving the adaptation of the model to different writing styles and cultural settings.

The handwriting samples were equally spaced over the five personality traits to preserve data balance and representativeness. Every document went through a preparation procedure involving noise reduction, feature extraction applied by automated image analysis methods, and scanning. This guaranteed the consistency of input data and reduced handwriting variances brought on by variances in writing devices, paper quality, or personal writing practices.

Particularly in situations with restricted data availability, the dataset was significantly enhanced to enhance the model generalisation. To strengthen the training set, augmentation methods call for rotation, scaling, and contrast modification. To help the efficient evaluation of the machine learning models, the last dataset was split into augmented training and test subsets.

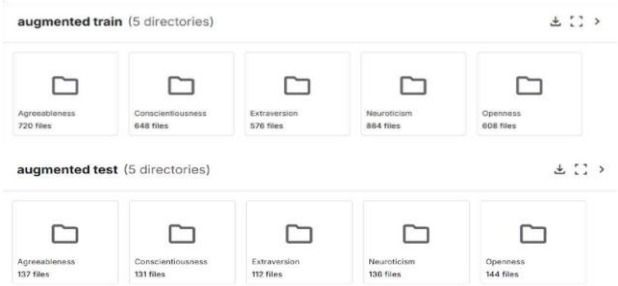



Fig.1 Dataset

Training and validation of the proposed handwriting-based personality prediction model were built upon this dataset. Its multilingual and varied character enables better generalisability across several scripts, writing styles, and linguistic origins, therefore enabling a major advance towards the development of an inclusive and flexible handwriting analysis system.

Predicting personality traits from handwriting is approached in many fundamental stages: dataset preparation, feature extraction, model design, training, and evaluation. Every stage was painstakingly created to guarantee great accuracy and efficiency in personality trait identification from handwriting characteristics.

Using a dataset comprising about 3,416 handwriting samples per personality trait—covering the Big Five Personality Traits—Openness, Conscientiousness, Agreeableness, Neuroticism, and Extraversion—this study examined To improve flexibility across linguistic and cultural settings, the dataset comprised samples from English and Hindi. Known to correlate with particular personality traits, each sample was examined for key handwriting characteristics including letter size, slant, pen pressure, word and letter spacing, and baseline alignment.

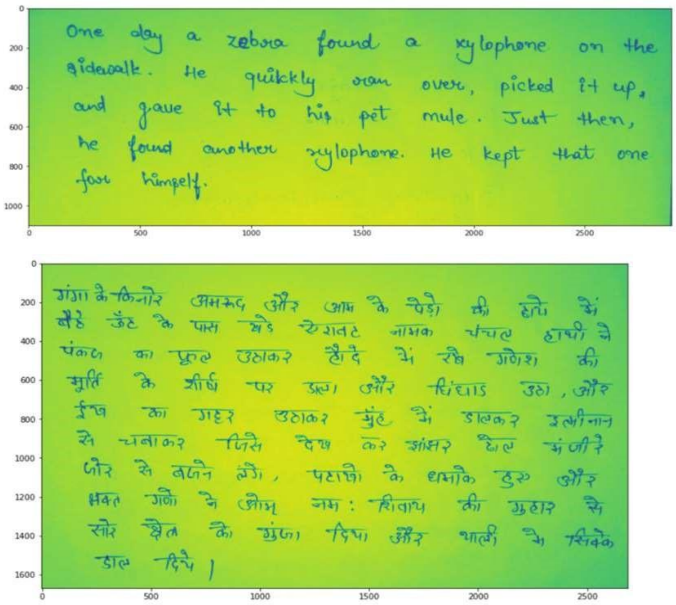



Fig. 2 Sample Dataset

Once collected, handwritten images undergo preprocessing to ensure standardization and consistency for effective feature extraction. The key preprocessing steps are as follows

Image Resizing: All handwriting samples were resized to maintain uniform dimensions, facilitating accurate feature extraction.

1. Feature Extraction: The following seven handwriting characteristics were analyzed, as they play a crucial role in personality identification.

2. Letter Size: Indicates confidence and focus; larger letters suggest extroversion, while smaller letters may reflect introversion or precision.
3. Writing Slant: Reveal emotional tendencies; a rightward slant is linked to expressiveness, while a leftward slant may in dicate introversion or emotional restraint.
4. Baseline Alignment: Shows mental state; an upward-sloping baseline suggests optimism, whereas a downward slope can signal pessimism or emotional strain.
5. Pen Pressure: Reflects emotional intensity; heavy pressure is associated with strong emotions, while light pressure suggests a more relaxed or passive disposition. Letter Spacing: Offers insight into social behavior; wider spacing suggests a preference for personal space, whereas narrow spacing indicates sociability.
6. Word Spacing: Similar to letter spacing, larger gaps between words indicate a preference for independence, whereas smaller gaps suggest a more social nature.
7. Top Margin: Provides clues about ambition and mindset; a larger top margin may indicate high aspirations, whereas a smaller margin suggests a more adaptable personality.

## V. Methodology

**Architectural System Development**

Multiple components working in a sequential manner to process handwriting input, extract essential handwriting features, classify personality traits, and generate job recommendations define the proposed system architecture for handwriting-based personality detection. Deep learning-based image categorisation is combined with conventional handwriting analysis and large language models (LLMs) for personality interpretation and job advice in a hybrid architectural style.

Handwriting Image Acquisition: Input Layer

The user uploads a handwritten text image at first in the system. Multiple steps of processing the input image help to extract pertinent information.

Two: Module for Preprocessing

Preprocessing the image guarantees accurate analysis and consists in:

The image is greyscale transformed to help in feature extraction simplicity.

Thresholding—adaptive or otsu—is used to distinguish text from the backdrop.

Unwanted artefacts are removed using morphological operations and median filtering.

3. Extraction of Handwritten Features

Image processing methods help to extract extra handwriting traits before feeding the image into the neural network. These characteristics consist of:

Average letter height is computed to assign handwriting either large, small, or average classification.

Handwriting direction is examined to ascertain if the text slants to the left, right, or remains vertical.

Pixel intensity values tell the writer whether she uses mild or heavy pressure.

Rising, declining, or irregular baselines are found by use of a polyn fitting method.

The average interval between words is calculated to ascertain whether words are spaced far apart or closely.

4. Depth Learning-Based Personality Classification

A VGG-16 deep neural network refined for personality trait classification feeds the processed image. One of the five Big Five personality traits—agreeableness, conscientiousness, extraversion, neuroticism, and openness—is predicted by the model. The classification relies on learnt patterns from handwriting traits.

5. Personality Interpretation Large Language Model (LLM)

A Large Language Model (LLM) is applied to offer a customised interpretation once the personality feature has been recognised. From this module:

a psychological research-based overview of the personality quality.

Understanding how the handwriting style either accentuates or balances with the expected characteristics.

Contextual knowledge on how such qualities affect performance at work.

6. System of Recommendation for Jobs

The solution lets HR experts upload job descriptions in order to offer useful applications of personality analysis. Based on the following parameters, the algorithm suggests the top three most fit professions using the categorised personality characteristic and LLM analysis:

harmony between job requirements and personal qualities.

handwriting traits implying pertinent strength.

LLM career insights tailored for a given industry.

7. Result and User Interaction

The result at last comprises:

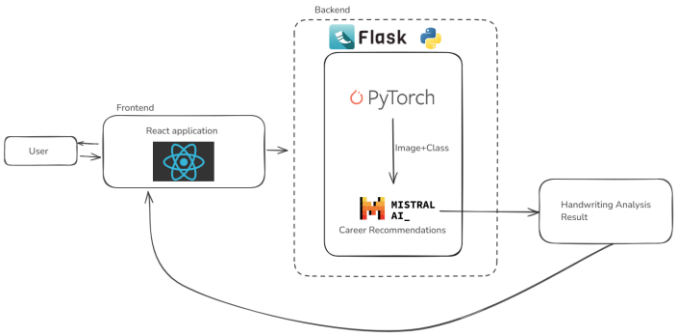
The expected quality of personality.

thorough examination of handwriting traits.

A career counselling report produced by artificial intelligence.

Top three professional suggestions.

This architecture presents a complete personality assessment and career advising system by efficiently combining deep learning, conventional handwriting analysis, and LLMs.




This work uses computer vision methods combined with analysis utilising a big language model to estimate personality traits derived from handwriting photos and provide customised career suggestions. Data preprocessing, model architecture design, training and evaluation of the model, feature extraction using handwriting analysis, and language model integration for summarising personality traits and offering counselling include various parts of the process.

Agreeableness, conscientiousness, extraversion, neuroticism, and openness are the five personality traits that the dataset comprises of categorised handwriting samples pertaining to. Directories for augmented training and testing were created using this information. Resized to 224x224 pixels, each handwriting image was normalised using standard means and standard deviations ([0.5, 0.5]) across all colour channels.

**Model Architecture**

Data augmentation methods were used offline prior to image loading in order to improve generalising capability. Foundation of our model was pre-trained ImageNet using the VGG16 convolutional neural network design. A fully linked layer with five output neurons—each corresponding to one of the Big Five personality traits—was added to the last classification layer. Only on the last few layers of the convolutional section was fine-tuning done to preserve low-level feature generalisation while enabling high-level changes.

We incorporated the Mistral Large Language Model (LLM) via Hugging Face to develop personalised summaries and offer career advice based on the discovered personality attribute, therefore improving the practicality and depth of our personality classification method. Once the CNN model built on VGG16 finds the dominant Big Five personality trait

Additional characteristics including slant, pen pressure, letter size, and word spacing are derived from handwriting photos. These features are then arranged for the LLM into a structured natural language prompt. Context is included in this query so that the LLM may more subtly interpret the psychological profile of the user. Using its vast pre-trained information, the LLM creates a personalised personality summary, makes career path recommendations, finds ideal workplace settings, and provides personal development advice. If someone is judged as high in "Openness" with large, right-slanted handwriting, for example, the LLM may suggest creative career paths such design, writing, or research while also suggesting techniques to help to further inspire inventive thinking.

This approach combines quantitative image analysis with qualitative, human-like interpretation to turn a simple categorisation task into a complete AI-based counselling tool, therefore rendering the results more meaningful and actionable for users. Using a CrossEntropy Loss function, an Adam optimiser with a learning rate of 0.001, and a StepLR learning rate scheduler to lower the learning rate after every five epochs, this setup was trained.

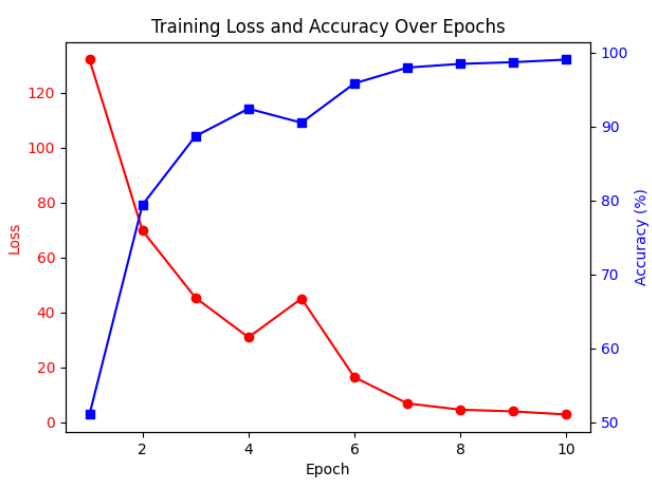
**Training and Evaluation**

The model underwent training for ten epochs using the training set with a batch size of 32. Each epoch included forward propagation, loss computation, backpropagation, and parameter updates. Upon completion of training, the model attained a testing accuracy of approximately 50% on previously unseen data, indicating its capability to generalize personality classification from handwriting images. The classification accuracy was determined by comparing the predicted labels with the ground truth labels from the test dataset. The model was preserved in PyTorch .pth format for future inference.

To enhance deep learning predictions, we also extracted handwriting-specific features utilizing OpenCV. Integration with Mistral LLM via Hugging Face After identifying the personality trait, a custom prompt was formulated and transmitted to the Mistral Large Language Model (LLM) through Hugging Face's API. The LLM was tasked to:

- Summarize the identified personality trait.
- Provide handwriting features as supporting evidence.
- Recommend personalized career paths, suitable work environments, and behavioral suggestions that correspond with the detected trait.
- The prompt included the personality label along with the handwriting features. The model returned a comprehensive, human-centered personality report and career recommendations.

## VI. Results



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